Chapter 4: Continuous Random Variables and Probability Distributions

PART B:

- UNIFORM DISTRIBUTION

Learning Objectives:

At the end of this chapter, the students would be able to:

- ✓ Understand the concept of a continuous random variable and probability distribution
- ✓ Be able to compute the expected value, variance, and standard deviation for a continuous random variable
- ✓ Be able to compute probabilities using a Uniform distribution, Normal distribution, Normal approximation to the Binomial distribution and Normal approximation to the Poisson distribution
- ✓ Continuous Probability Distributions (Probability Mass Functions, Mathematical Expectations)
- ✓ Special Continuous Probability Distributions (Uniform Normal; Normal Approximation to the Binomial Distribution; Normal Approximation to the Poisson Distribution)

4.2 Special CRV: Uniform Distribution

The density function of the continuous uniform random variable X on the interval [A, B] is:

$$f(x; A,B) = \begin{cases} \dfrac{1}{B-A}, & A \le x \le B \\ 0, & \text{elsewhere} \end{cases}$$

A = lower limit
B = upper limit

The uniform distribution is often called the rectangular distribution because the density function forms a rectangle with base B–A and constant height $\frac{1}{B-A}$. The interval may not always be closed [A, B]. It can be (A, B) as well.

4.2 Special CRV: Uniform Distribution

The mean and variance of the uniform distribution are:

$$\mu = \frac{A + B}{2}$$

$$\sigma^2 = \frac{(B - A)^2}{12}$$

4.2 Special CRV: Uniform Distribution

- Types of continuous probability distribution:

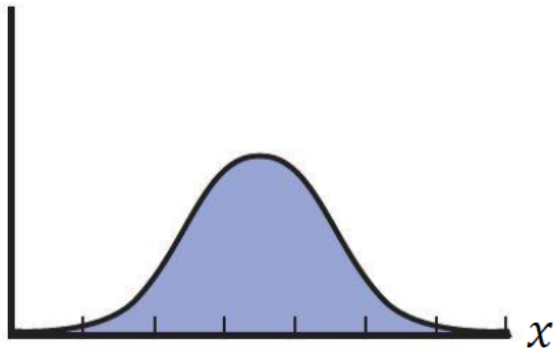


Figure 1:
Normal distribution

- Mean = Median
- $-\infty < X < \infty$

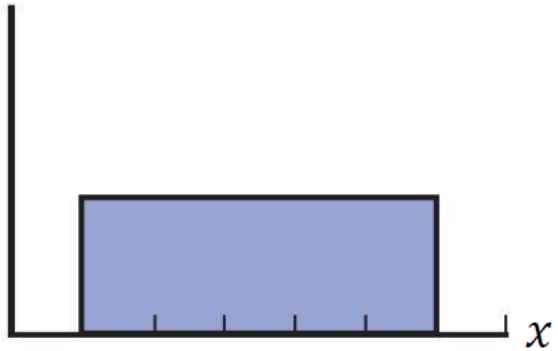


Figure 2:
Uniform distribution

- Mean = Median
- $a < X < b$

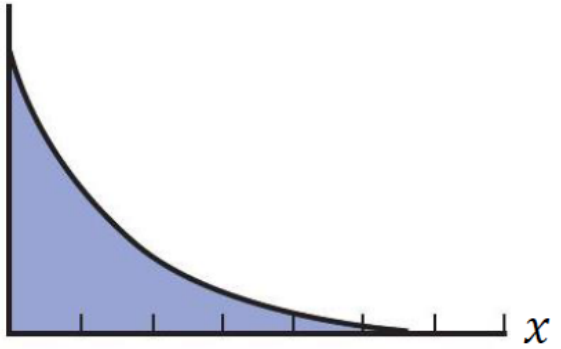


Figure 3:
Exponential distribution

- Mean > Median
- $0 < X < \infty$

4.2 Special CRV: Uniform Distribution

Example 1 (from textbook Introduction to Probability and Statistics, UUM, (2021); pg. 89)

You arrive into a construction and are about to take an elevator to your floor. Once you call the elevator, it will take between 0 and 30 seconds to arrive to you. We will assume that the elevator arrives uniformly between 0 and 30 seconds after you press the button. Find the **probability density function** and **mean**.